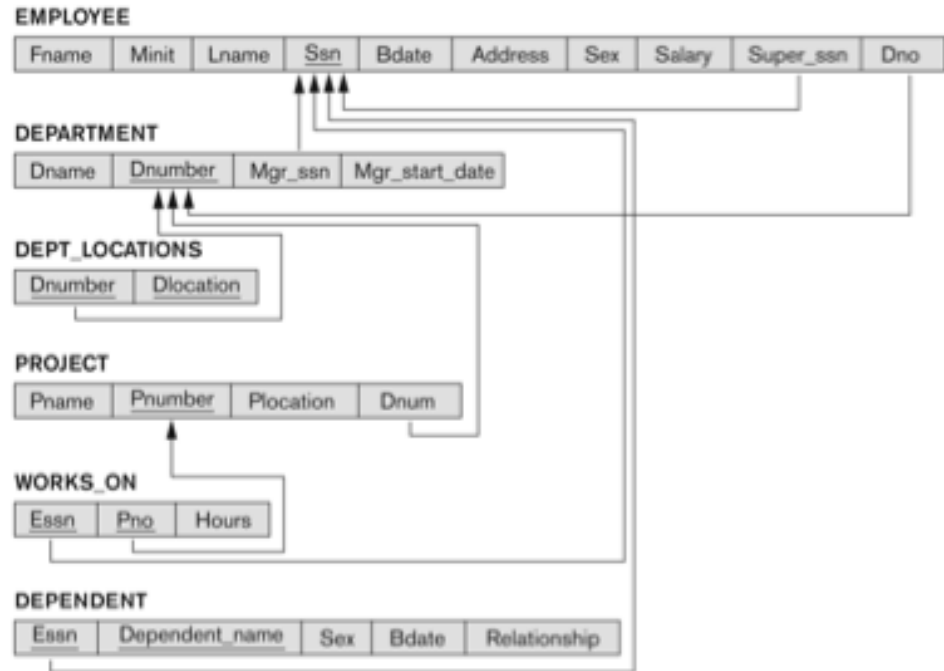


أهلاً .. حيث أقولكم على اختبار الداتا كيف كان
هو بدايةً كأنه سهل .. محتاج انه مذاكر SQL كويس
جداً لئلا لا يخيار كان فيه كل الاقل ١٥ أسئلة عند
SQL .. بعضها سهل وواحد من جدا وبعضها كانت فيها
فكرة اجبت شوية بس تخط يعني ..
* ضروري تتدربوا بنفسكم كويس ..

re.

Figure 5.7

Referential integrity constraints displayed on the COMPANY relational database schema.



الأسئلة جات على هذا

حار 3 اسئلة على Algebra

كانوا واجهين

جاء سؤال mapping كان حول بس واضح

خبره تكونوا فاهمينه في سوالها جابت
حالات كثيرة

جاء جمع وخطا من ال جزاء المقاليه

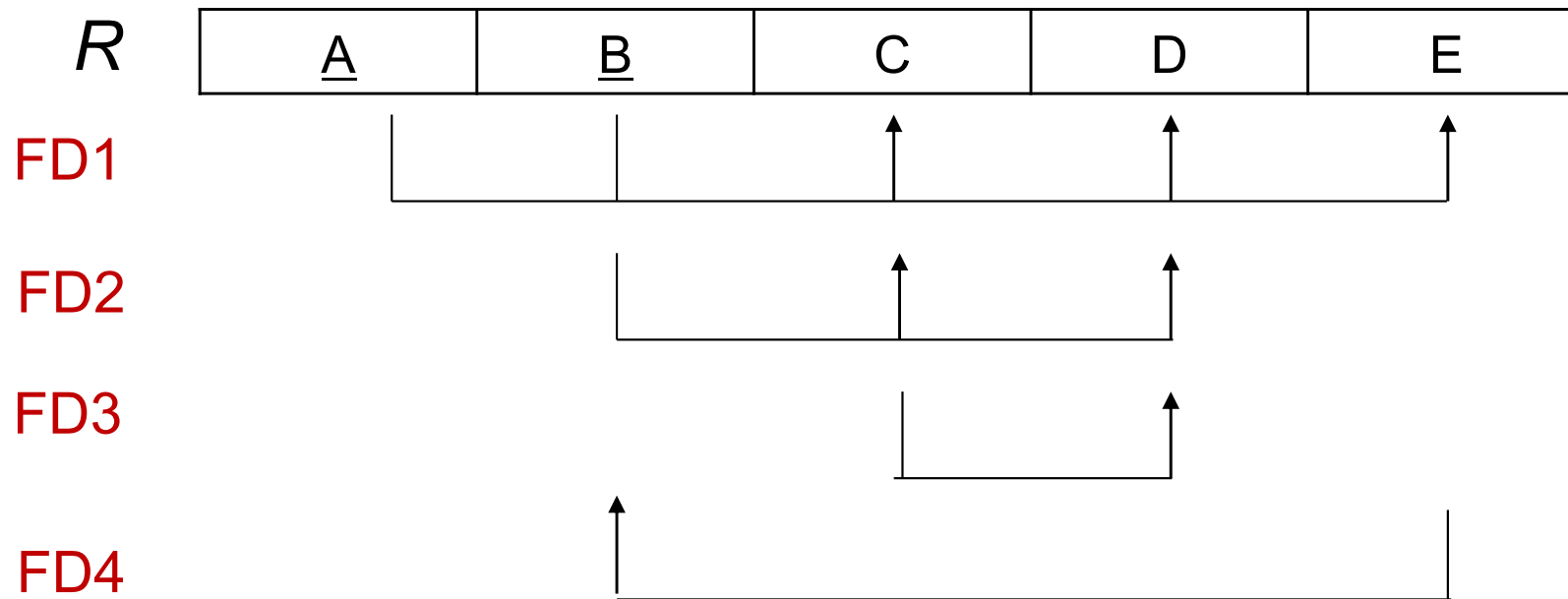
كانسها مرة

جاء سؤال على normal جدول

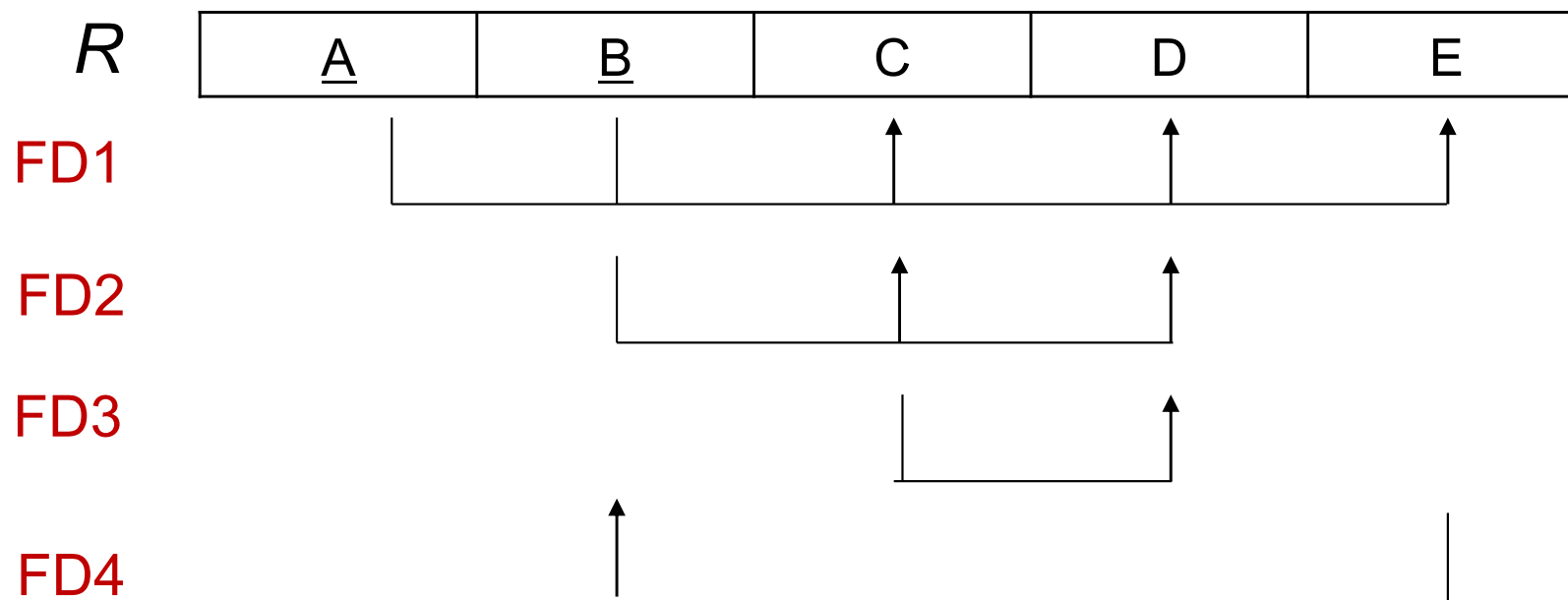
وتشوف أي normal ما هو متحقق
form

تست نرمالیت برای رابطه R ؟

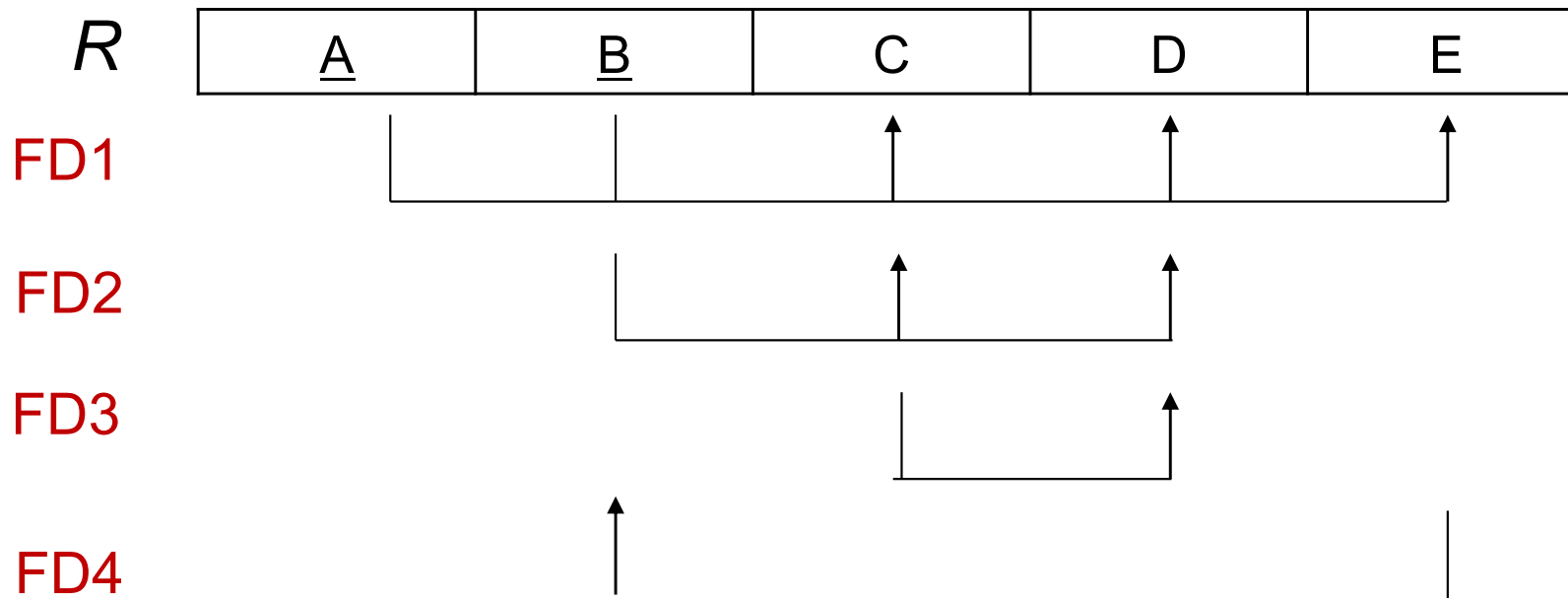
Test Normality for the relation R ?



Is it in the 1NF ??

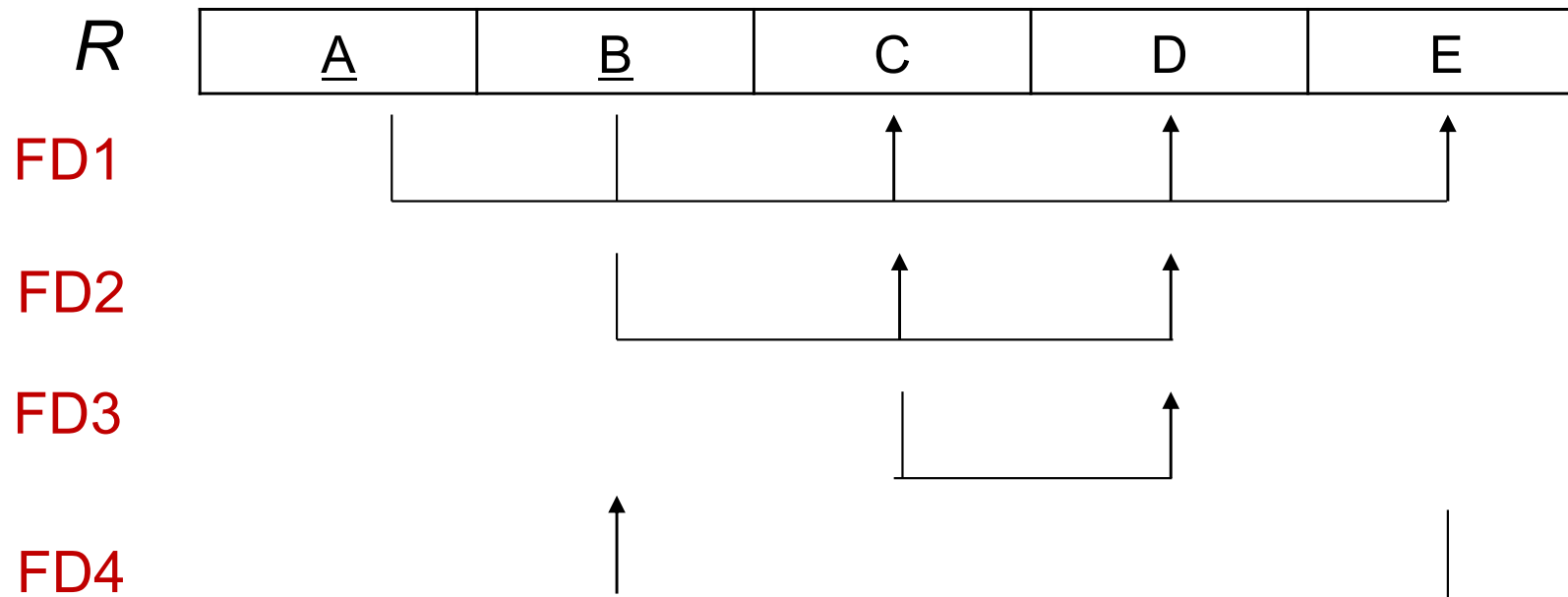


Is it in the 1NF ??

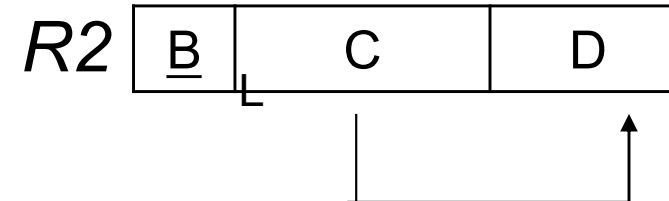
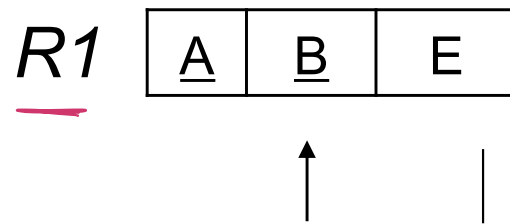
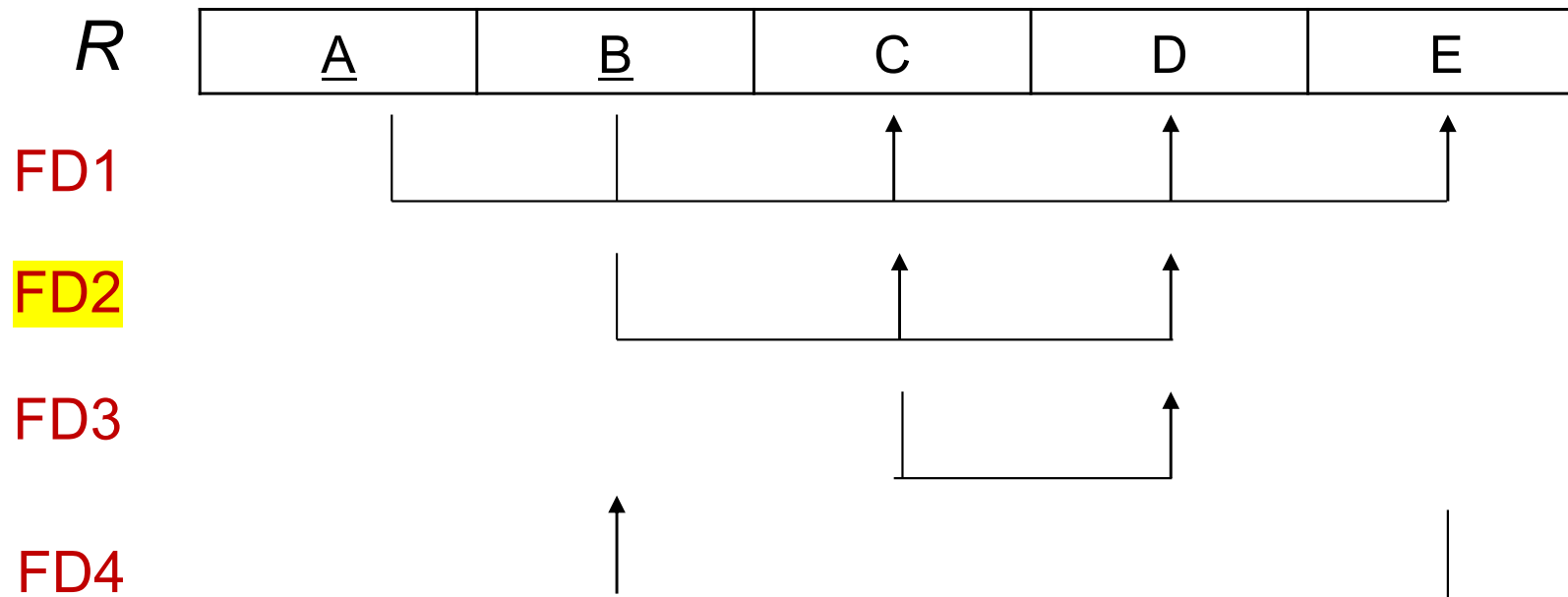


Yes, R is in the 1NF because there is no violation of 1NF.

Is it the 2NF ??

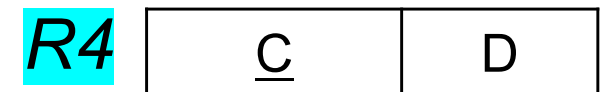
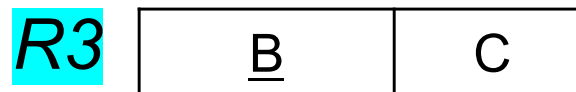
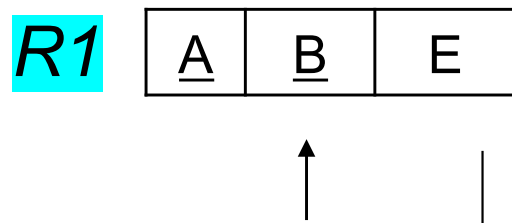
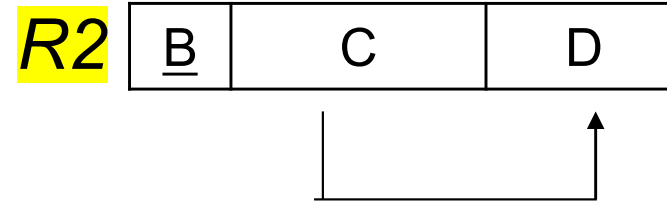
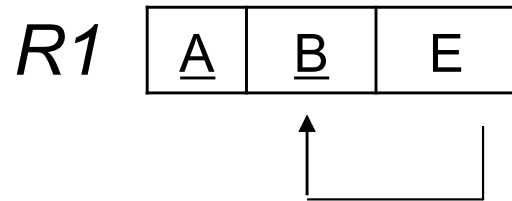


Is it the 2NF ??

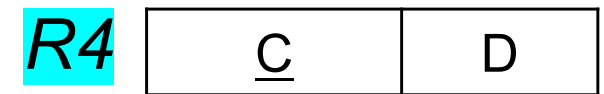
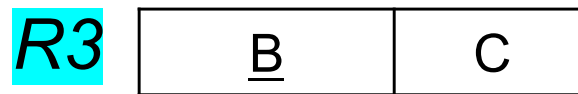
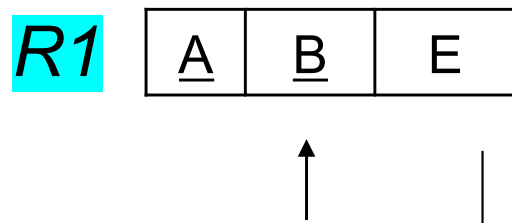
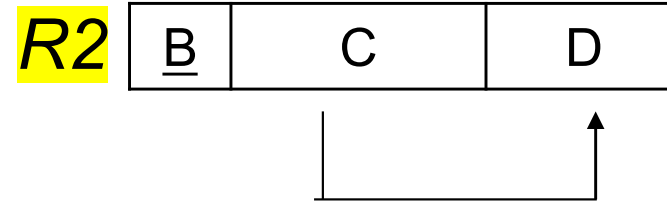
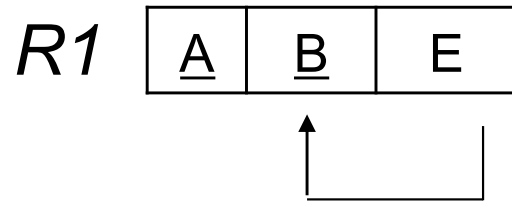


Is it the 3NF ??

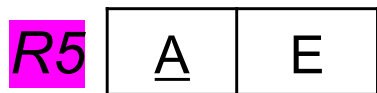
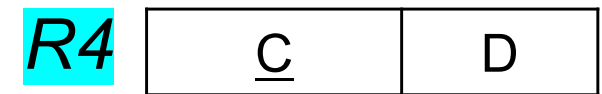
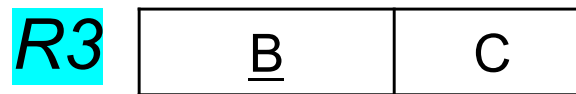
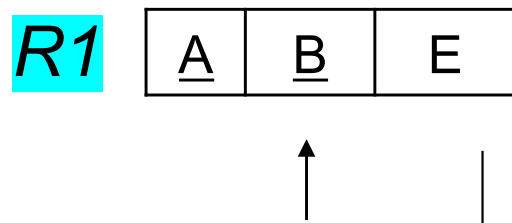
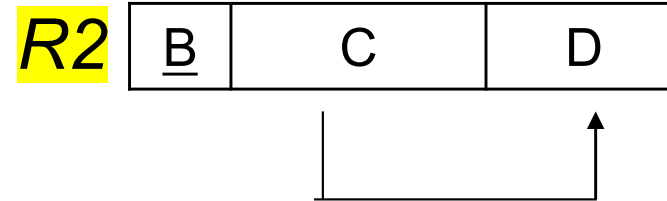
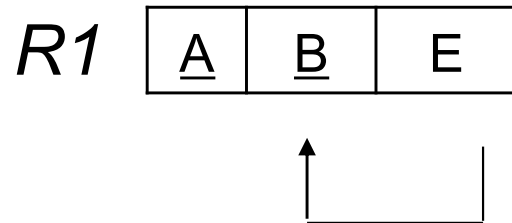
C



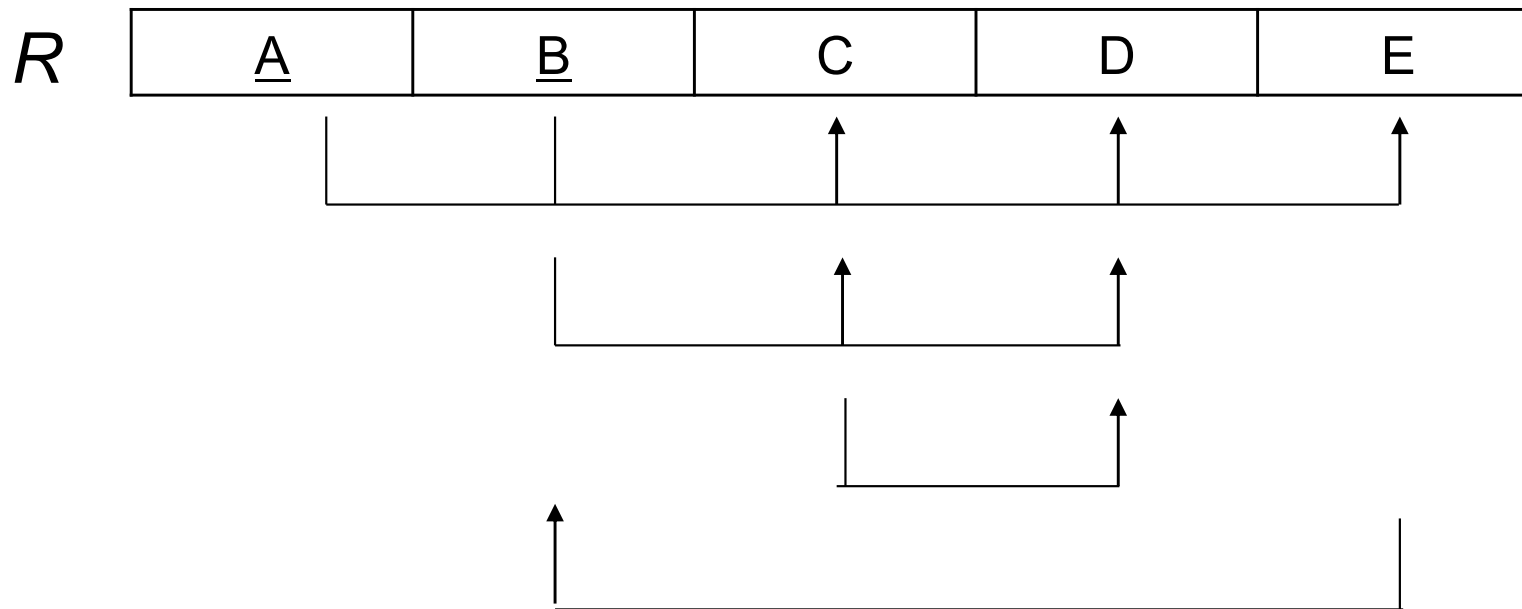
Is it the BCNF ??



Is it the BCNF ??



The Solution



R5

<u>A</u>	E
----------	---

R6

<u>E</u>	B
----------	---

R3

<u>B</u>	C
----------	---

R4

<u>C</u>	D
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حالة فصي هذا السؤال

Normalization Exercise 2

7

	INVOICE 2 HILLTOP ANIMAL HOSPITAL INVOICE # 987 1 DATE: JAN 13/2002																		
3																			
4	MR. RICHARD COOK 123 THIS STREET MY CITY, ONTARIO Z5Z 6G6																		
5	<table><tr><th>PET</th><th>PROCEDURE</th><th>AMOUNT</th></tr><tr><td>ROVER</td><td>RABIES VACCINATION</td><td>30.00</td></tr><tr><td>MORRIS</td><td>RABIES VACCINATION</td><td>24.00</td></tr><tr><td colspan="2">TOTAL</td><td>54.00</td></tr><tr><td colspan="2">TAX (8%)</td><td>4.32</td></tr><tr><td colspan="2">AMOUNT OWING</td><td>58.32</td></tr></table>	PET	PROCEDURE	AMOUNT	ROVER	RABIES VACCINATION	30.00	MORRIS	RABIES VACCINATION	24.00	TOTAL		54.00	TAX (8%)		4.32	AMOUNT OWING		58.32
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Normalization Exercise 2

<u>INVOICE</u>		
HILLTOP ANIMAL HOSPITAL INVOICE # 987		DATE: JAN 13/2002
MR. RICHARD COOK		
123 THIS STREET MY CITY, ONTARIO Z5Z 6G6		
<u>PET</u>	<u>PROCEDURE</u>	<u>AMOUNT</u>
ROVER	RABIES VACCINATION	30.00
MORRIS	RABIES VACCINATION	24.00
TOTAL		54.00
TAX (8%)		<u>4.32</u>
AMOUNT OWING		<u>58.32</u>

Normalization Exercise 2 - Solution

- Unnormalized form (UNF):

- invoice [invoice_no, invoice_date, cust_name, cust_addr, pet_name, procedure, amount]

Normalization Exercise 2 - Solution

- Unnormalized form (UNF):
 - invoice [invoice_no, invoice_date, cust_name, cust_addr, pet_name, procedure, amount]
- Is it in the 1NF ??

Normalization Exercise 2 - Solution

- Unnormalized form (UNF):
 - invoice [invoice_no, invoice_date, cust_name, cust_addr, pet_name, procedure, amount]
- 1NF:
 - invoice [invoice_no, invoice_date, cust_name, cust_addr]
 - ✗ invoice_pet [invoice_no, pet_id, pet_name, procedure, amount]

Normalization Exercise 2 - Solution

■ 1NF:

- **invoice** [invoice_no, invoice_date, cust_name, cust_addr]
- **invoice_pet** [invoice_no, pet_id, pet_name, procedure, amount]



Is it in 2NF ?

Normalization Exercise 2 - Solution

■ 1NF:

- invoice [invoice_no, invoice_date, cust_name, cust_addr]
 - invoice_pet [invoice_no, pet_id, pet_name, procedure, amount]
- 

■ 2NF:

- invoice [invoice_no, invoice_date, cust_name, cust_addr]
- invoice_pet [invoice_no, pet_id, procedure, amount]
- pet [pet_id, pet_name]

Normalization Exercise 2 - Solution

■ 2NF:

- invoice [invoice_no, invoice_date, cust_name, cust_addr]
- invoice_pet [invoice_no, pet_id, procedure, amount]
- pet [pet_id, pet_name]



Is it in 3NF ?

Normalization Exercise 2 - Solution

■ 2NF:

- invoice [invoice_no, invoice_date, cust_name, cust_addr]
- invoice_pet [invoice_no, pet_id, procedure, amount]
- pet [pet_id, pet_name]



■ 3NF:

- invoice [invoice_no, invoice_date, cust_name, cust_addr]
- invoice_pet [invoice_no, pet_id, procedure]
- invoice_procedure [procedure, amount]
- pet [pet_id, pet_name]